On Sense and Reference

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PHI 205

This is a reading guide. It tells you what the assigned reading is, where to find it, and offers some guidance on how to read it.

This is our first "real" reading assignment, so I will be giving a lot of guidance.

Getting the Paper

The assigned reading is Frege (1948) , which is a translation of Frege (1892). If you read German, obviously you should read the original.

The link I have given you to Frege (1948) is a "DOI" link. "DOI" stands for "Digital Object Identifier." When you click on the link, it will take you to JSTOR, a large repository of journal articles.

Access to the article is not free, but our library has a subscription. If you are accessing the article off campus, you will need to click on the link to "Login through your library".

Downloading and Printing the Paper

You might be tempted to just read the article online. But you should instead save the PDF to your device, and print out a copy. You should create a folder on your device for this class and save all the PDFs in that folder.

You should print the paper out because, on our in-class writing days, you will not be allowed to use electronic devices, but you are allowed to use print-outs of the articles and your notes. Print the article now, and read the hard-copy, making notes in the margins. Bring it to class so you can follow along as we discuss the article, and take more notes. That way, when the writing day rolls along, you will have a copy of the paper along with a written record of your interaction with it.

Reading Assignment

I am **not assigning the entire paper**. You can skip from p. 218, the paragraph that begins, "We are thus led to consider subordinate sentences or clauses," to the end of the paper. I do want you to read the last two paragraphs, on p. 230, beginning with "Let us return to our starting point!"

We are skipping the 12 page discussion of subordinate clauses. Of course you are welcome to read it. It is interesting. But it is not assigned.

How to Read

You already know how to read. And most of you have already taken philosophy classes, so you already know how to read philosophy.

But this is a tough paper. Generations of philosophy students have cut their teeth on it, and it is worth reading slowly and carefully. Below, I guide you through the first two pages, sentence by sentence, before setting you free to read the rest of it on your own.

Before we get started, some advice:

Give yourself grace. You will be confused. It will be frustrating. You might have trouble caring about it. It is okay to take breaks, but remember, this is what you signed up for, and realize that it is not impossible. As mentioned above, generations of philosophy students have done this before you, and gotten a lot out of it.

As you read, **always be trying to figure out what is going on**. With difficult reading, it is easy to fall into a passive mode, letting the words wash past you. At best, this lets you get a vibe from the article. It might give you the feeling that you understand some of it, but it will not give you actual understanding. To actually understand, you need to work. Check your understanding by trying to apply examples—your own or examples from the article—to abstract claims. Constantly ask yourself how the different claims the author makes are supposed to fit together: is this claim meant as evidence for that other one? As an objection? Is this example an illustration of that principle? A counterexample?

As you do this, you need to take notes. "Taking notes" is not about highlighting important passages—that is almost useless, and if you are in the habit of doing it, you should try to break that habit. It is also not about recording key definitions or facts. Your notes should be a written record of your struggle with the text: **jot down** your confusion, your examples, your objections, your uncertainty, your doubts, your moments of understanding, moments of agreement, and moments of disagreement. These are the notes you need. The process of writing them down helps to solidify your own understanding. When you return to the text, they will help you remember where your head was at last time, so you can build on that.

Argue with the paper as you read it. Your goal is not just to understand it. Your goal is to figure out whether or not you agree with it. Try to identify key premises, and then try to come up for reasons for thinking they might be false. Keep your eye out for places where the author says something is "clear" or "obvious" or "natural", and ask yourself whether it really is.

I said above that you need to **read slowly and carefully**. If you are always trying to figure out what is going on, always arguing with the paper, and constantly jotting all of this down, it is going to take you a long time to get through the text. Don't rush the process. Let it take the time it needs to take. Notice that there are no out-of-class writing assignments for this class. I expect you to use that extra time reading and re-reading, thinking and re-thinking.

Another common piece of advice: read the first paragraph or two, then read the last paragraph or two. After doing this, try to figure out what you think is going to happen in the rest of the paper. Read the rest of the paper, and see if it matches your expectations or if it surprises you.

Another common piece of advice: ignore what I said above about reading slowly and carefully. Blast your way through the article as quickly as possible, giving your brain a first exposure, vibing with it as best you can, without worrying too much about how it all fits together. **Then** read it a second time through, slowly and carefully.

A final universal piece of advice: don't expect to get everything you need to get out of the paper on the first reading or even second reading. Your first reading puts you in a position to *begin* to understand what is going on. Lecture and discussion to re-read the paper and understand it better. After reading other authors on related topics, you should return to the paper and read it yet again, and you will understand it yet better. With rich philosophical works, this process can go on forever.

Enough advice. Let's get reading.

Reading Notes

I am going to walk you through the first page and a half, sentence by sentence. I won't be doing this for every reading. But we are just getting started, and this reading is dense, and I want to give you a model of what I mean by "read slowly and carefully."

> Identity[1] gives rise to challenging questions which are not easy to answer.
>
> [1] I use this word strictly and understand "$a = b$" to have the sense of "a is the same as b" or "a and b coincide."

What do you think Frege means by 'identity'? The fact that he provides a footnote suggests that he was worried that you might misunderstand.

> Is it a relation?

Here are some examples of relations:

- The *greater than* relation: '$4 > 3$', $4 > 5$.
- The *worse than* relation: 'This pasta is *worse than* the pasta you made last night.'
- The *is a part of* relation: 'My nose *is a part of* my face.'

What are some other examples of relations? Here are some identity claims:

- $2 + 2 = 4$
- Dwayne Johnson *is identical to* the Rock.
- Daniel Radcliffe *is identical to* Elijah Wood.

Can you think of reasons someone might have for thinking that identity *is not* a relation?

> A relation between objects, names or signs of objects?

Huh. What are these two options? Why are they the two options we should consider?

> In my *Begriffsschrift* I assumed the latter.

The *Begriffsschrift* ("Concept Notation") (Frege 1879; English translation Frege 1967) is a booklet on logic. In it, Frege invents the logic that we now teach in intro logic courses, displacing traditional Aristotelian term logic.

In §8 of that booklet, "Identity of content", he says,

> Whereas in other contexts signs are merely representatives of their content […] they suddenly display their own selves when they are combined by means of the sign for identity of content; for it expresses the circumstance that two names have the same content. (Frege 1967, 20)

Returning to our reading, he continues,

> The reasons which seem to favor this are the following:

Ah, good. We won't need to read the *Begriffsschrift*, because he is about to explain his reasons to us here. (Notice the use of 'seem'. That suggests he has changed his mind.)

> $a = a$ and $a = b$ are obviously statements of differing cognitive value; …

Note that Frege does not follow our contemporary convention for use-mention. Following this convention, we would surround the two identity statements, which are here being mentioned, not used, in single quotes:

> '$a = a$' and '$a = b$' are obviously statements of differing cognitive value; …

I will not be "fixing" Frege's use-mention to fit this convention. But you should pay attention to it for yourself.

He does not explain what he means by "cognitive value". "Cognitive" means, roughly, "related to thought". So "thought value"? Maybe this will be clearer in light of the reason he offers in support:

> $a = a$ holds *a priori* and, according to Kant, is to be labeled analytic, while statements of the form $a = b$ often contain very valuable extensions of our knowledge and cannot always be established *a priori*.

Here Frege uses some philosophical terminology and name-drops Kant. He assumes anyone reading this article has already read Kant ([1781] 1998). He's probably wrong about that in your case. You should look up what '*a priori*' and 'analytic' mean (hint: here 'analytic' is being used in contrast to 'synthetic'). And you should put Kant ([1781] 1998) on your summer reading list!

Did you look up what '*a priori*' and 'analytic' mean? Do so before reading further.

Now you know that Frege is claiming that statements of the form '$a = a$' can be known independently of experience (that is, "hold *a priori*"), and are, at least according to Kant, true in virtue of the meanings of the words and how they have been put together (that is, "analytic"), while statements of the form '$a = b$' are often both synthetic and *a posteriori*—in order to know that $a = b$, you need to have empirical evidence, not just know what the words mean and how they have been put together.

Does all of that seem right to you? Try to think about specific examples. Do you agree that,

- Dwayne Johnson = Dwayne Johnson,

is both *a priori* and analytic? Do you agree that,

- Dwayne Johnson = the Rock.

is neither analytic nor *a priori*?

> The discovery that the rising sun is not new every morning, but always the same, was of very great consequence for astronomy. Even today, the identification of a small planet or a comet is not always a matter of course.

That is, the statement,

- The sun that rises today = the sun that rose yesterday,

is neither analytic nor *a priori*, but instead expresses an empirical scientific discovery.

Let's stop to recap. Reading from the top:

> Identity gives rise to challenging questions which are not easy to answer. Is it a relation? A relation between objects, names or signs of objects? In my *Begriffsschrift* I assumed the latter. The reasons which seem to favor this are the following: $a = a$ and $a = b$ are obviously statements of differing cognitive value; … $a = a$ holds *a priori* and, according to Kant, is to be labeled analytic, while statements of the form $a = b$ often contain very valuable extensions of our knowledge and cannot always be established *a priori*. The discovery that the rising sun is not new every morning, but always the same, was of very great consequence for astronomy. Even today, the identification of a small planet or a comet is not always a matter of course.

See where we are? Frege has asserted that identity is either a relation between objects or between names of objects, and he is in the middle of giving us a reason for thinking it is not a relation between objects. So far, he has argued that '$a = a$' and '$a = b$' differ in cognitive value. But it remains to be seen why that should entail that identity is not a relation between objects.

> Now if we were to regard identity as a relation between that which the names "a" and "b" designate, it would seem that $a = b$ could not differ from $a = a$ (i.e., provided $a = b$ is true). A relation would thereby be expressed of a thing to itself, and indeed one in which each thing stands to itself but to no other thing.

Frege's argument here is that identity cannot be a relation between objects, because, if it were, '$a = a$' and '$a = b$' would have the same cognitive value. Why? Because both '$a = a$' and '$a = b$' would express the same relation—a relation that holds between a thing and itself, and not between that thing and anything else.

(Notice that this helps clarify the kind of 'identity' Frege is interested in here. *Qualitative identity*—the kind of identity-as-similarity that holds between "identical" twins—is a relation that can hold between one thing and another thing. *Numerical identity*—being one and the same thing—is the relation that holds between each thing and itself, but not between that thing and anything else.)

We have been given an argument. It is good practice, when reading philosophy, to try to reconstruct the arguments you are given.

- (NO1) '$a = a$' and '$a = b$' differ in cognitive value.
- (NO2) If identity were a relation between objects, '$a = a$' and '$a = b$' would express the same relation.
- (NO3) If '$a = a$' and '$a = b$' expressed the same relation, they would have the same cognitive value.
- (NO4) So, identity is not a relation between objects.

It is also good practice to immediately start trying to think of objections. This isn't because we are playing some obnoxious game of "Gotcha!" It's because, by trying think of objections, you force yourself to actually pay attention to what is being said. Students often read philosophy in "vibe" mode, and come away with a feeling of understanding, but no actual understanding. You have to fight this tendency.

So, what do you think? Is the argument just reconstructed valid? That is, if you were to grant the premises, would the conclusion follow? Assuming it is valid, are there any premises that look fishy to you? If not,

imagine you are a really disagreeable reader, searching for things to disagree with. Where might you poke or prod?

If you just tried to do all of that, you may have run into the problem that the argument is pretty abstract, so it is hard to get real purchase on it. When that happens to you, always try to reach for examples. I've given you the 'Dwayne Johnson' and 'The Rock' examples. Try plugging those into the argument, in place of the abstract '$a = a$' and '$a = b$'.

Also, it is fine if you don't come up with any great objections. The point is to try. The goal isn't a "Gotcha!" The goal is to actually understand the argument.

> What is intended to be said by $a = b$ seems to be that the signs or names "a" and "b" designate the same thing, so that those signs themselves would be under discussion; a relation between them would be asserted.

Ah, now we are moving on to the second option: if identity is not a relation between objects, it must be a relation between names. What relation, exactly? Co-reference: when you learn that Dwayne Johnson is the Rock, what you have learned is that the name, 'Dwayne Johnson', and the name, 'The Rock', both refer to the same thing. This is a relation between the two names: they co-refer.

> But this relation would hold between the names or signs only insofar as they named or designated something. It would be mediated by the connection of each of the two signs with the same designated thing. But this is arbitrary. Nobody can be forbidden to use any arbitrarily producible event or object as a sign for something. In that case the sentence $a = b$ would no longer refer to the subject matter, but only to its mode of designation; we would express no proper knowledge by its means. But in many cases this is just what we want to do.

Oh, shit. Now we seem to have an argument against the view that identity is a relation between names. Let's try to reconstruct it.

- (NN1) The connection between a name and its referent is arbitrary and conventional.
- (NN2) So, the fact that 'a' and 'b' co-refer is arbitrary and conventional.
- (NN3) But sentences of the form '$a = b$' can express proper knowledge.
- (NN4) Knowledge of arbitrary and conventional facts is never proper knowledge.
- (NN5) So, sentences of the form '$a = b$' do not express co-reference.

I've done a fair bit of work for you here, unpacking the argument from the text. I may have gotten it wrong. So, once again, you should be thinking of possible objections, and thinking about which premise(s) seem most dubious. But also, you should be comparing what I've given you with the original text, and thinking about whether my reconstruction is misleading.

> If the sign "a" is distinguished from the sign "b" only as object (here, by means of its shape), not as sign (i.e., not by the manner in which it designates something), the cognitive value of $a = a$ becomes essentially equal to that of $a = b$, provided $a = b$ is true.

This sentence is hard to parse. It helps a little if we move the last bit, 'provided $a = b$ is true', to the beginning:

- If '$a = b$' is true, and if each name is distinguished from the other only by its shape, not by the "manner in which it designates something", then '$a = b$' has the essentially the same cognitive value as '$a = a$'.

Frege's first argument suggested that, when $a = b$ is true, the relation between the object a and the object b is the same as the relation between the object a and the object a, so nothing at the level of objects can explain a difference in cognitive value.

His second argument suggests that the fact that two signs of two different shapes have both been connected to the same object is arbitrary and conventional. That gets us a difference, but not the right sort of difference. The difference we are after needs to be a difference that can amount to proper knowledge: again, think of the case of the astronomer who discovers that comet observed 100 years ago is identical to the comet she observes today.

> A difference can arise only if the difference between the signs corresponds to a difference in the mode of presentation of that which is designated.

Here, Frege introduces a third potential source of a difference in cognitive value: not the difference in names, and not the difference in objects, but the difference in "modes of presentation" by which each name designates the object.

But what is a "mode of presentation"? What follows is an example, illustrating the idea:

> Let a, b, c be the lines connecting the vertices of a triangle with the midpoints of the opposite sides. The point of intersection of a and b is then the same as the point of intersection of b and c. So we have different designations for the same point, and these names ("Point of intersection of a and b," "Point of intersection of b and c") likewise indicate the mode of presentation; and hence the statement contains true knowledge.

Construct a diagram to help yourself make sense of this example.

> It is natural, now, to think of there being connected with a sign (name, combination of words, letter), besides that to which the sign refers, which may be called the referent of the sign, also what I would like to call the *sense* of the sign, wherein the mode of presentation is contained. In our example, accordingly, the referents of the expressions "the point of intersection of a and b" and "the point of intersection of b and c" would be the same, but not their senses. The referent of "evening star" would be the same as that of "morning star," but not the sense.

This is Frege's famous view about how names work: each name *expresses* a sense, which *determines* a referent. The difference in cognitive value between 'Dwayne Johnson is Dwayne Johnson' and 'Dwayne Johnson is the Rock' is due to the fact that 'Dwayne Johnson' and 'The Rock' each express a different sense —a different "mode of presentation"—even though those two different senses determine the same object.

Here ends this reading guide, but not your reading assignment. You still have several pages to go. In those pages, Frege will clarify his view that He then extends that view from names to entire sentences: each sentence *expresses* a sense, which *determines* a referent. In the case of a sentence, the sense is a "though", and the referent is a truth value. And he discusses what happens when a name has a sense but lacks a referent, and how that affects the sense and reference of a sentence containing that name.

Try to use this guide as a model. Read slowly. Take each sentence as it comes, and try to figure out what it is saying. Obviously, in writing this guide, I was cheating. I've read this paper many times, so I already know what is going on. You are reading this paper for the first time. You will have to read ahead and circle back, and you won't always be able to make sense of what is going on. That's all okay! Stick with it and push through. You cannot understand a paper because you've read it several times without reading it for the first time, without completely understanding it.

Slides

I used slides for our second day discussing Frege. Here are those slides.

References

Frege, G. 1967. "Begriffsschrift, a Formula Language, Modeled Upon That of Arithmetic, for Pure Thought." In *From Frege to Gödel*, edited by Jean van Heijenoort. Harvard University Press.

Frege, Gottlob. 1892. "Über Sinn Und Bedeutung." *Zeitschrift Für Philosophie Und Philosophische Kritik* 100 (1): 25–50. https://www.deutschestextarchiv.de/book/show/frege_sinn_1892.

Frege, Gottlob. 1948. "Sense and Reference." Translated by Peter Geach. *The Philosophical Review* 57 (3): 209–30. https://doi.org/10.2307/2181485.

Frege, Gottlob. 1879. *Begriffsschrift, Eine Der Arithmetischen Nachgebildete Formelsprache Des Reinen Denkens*. Nebert.

Kant, Immanuel. (1781) 1998. *Critique of Pure Reason*. Translated by Paul Guyer and Allen W. Wood. Cambridge University Press.